Solutions to:
From Newton to Navier-Stokes

Section 3

3.1)

(a) $\vec{x}(t) = \vec{x}_0 + \vec{\psi}(\vec{x}_0, t)$

(b) Differentiating gives

$$\vec{\mathrm{v}}(t) = \frac{d\vec{x}}{dt} = \frac{d\vec{\psi}}{dt}(\vec{x}_0, t)$$

3.2) In order for both descriptions to be equivalent, the instantaneous velocity at a given position and a given time should agree. However, this means the position we plug into the Eulerian velocity field also depends on time, since the position of a given fluid element depends on time. Thus, we should have

$$\vec{\mathrm{v}}(t) = \vec{v}\,[\vec{x}(t), t]$$

Then, using the results from the previous exercise, we get

$$\frac{d\vec{\psi}}{dt}(\vec{x}_0, t) = \vec{v}\left[\vec{x}_0 + \vec{\psi}(\vec{x}_0, t), t\right]$$

Section 4

4.1) Here we track a specific fluid element over time, so the position also depends on time. Thus, chain rule gives

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial x}\frac{dx}{dt} + \frac{\partial \phi}{\partial y}\frac{dy}{dt} + \frac{\partial \phi}{\partial z}\frac{dz}{dt} = \frac{\partial \phi}{\partial t} + \nabla\phi \cdot \vec{v} = \frac{\partial \phi}{\partial t} + (\vec{v}\cdot\nabla)\phi$$

Using index notation gives a slightly more compact way of writing chain rule:

$$\frac{d\phi}{dt} = \frac{\partial \phi}{\partial t} + \frac{\partial \phi}{\partial x_i}\frac{dx_i}{dt} = \frac{\partial \phi}{\partial t} + \left(\frac{dx_i}{dt}\frac{\partial}{\partial x_i}\right)\phi = (\vec{v}\cdot\nabla)\phi$$

We write $\vec{v}\cdot\nabla$ to avoid ambiguity, which will become clear when we start dealing with vector fields.

When we take a partial derivative w.r.t. t, we keep the spatial coordinates x, y, z fixed, but we have to remember the fluid element itself also moves over time, so after a small time δt, the fluid element at the original position may not be the same one. Thus, the term with the gradient dotted into the velocity accounts for this change in position.

4.2)

(a) From *Exercise 3.2*, we have

$$\frac{d}{dt}\vec{\mathrm{v}}(t) = \frac{d}{dt}\vec{v}\,[\vec{x}(t), t] = \frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}$$

(b) Let us separate out the components of position, so

$$\vec{v}\,[\vec{x} + \delta\vec{x}, t + \delta t] = \vec{v}\,[x + \delta x, y + \delta y, z + \delta z, t + \delta t]$$

where $\delta\vec{x} = \vec{v}\,\delta t = (v_x\,\delta t)\hat{\mathbf{x}} + (v_y\,\delta t)\hat{\mathbf{y}} + (v_z\,\delta t)\hat{\mathbf{z}}$

Doing a first-order Taylor expansion gives

$$\begin{aligned}
\vec{v}\,[x+\delta x, y+\delta y, z+\delta z, t+\delta t] &\approx \vec{v}\,[x,y,z,t] + \frac{\partial \vec{v}}{\partial x}\delta x + \frac{\partial \vec{v}}{\partial y}\delta y + \frac{\partial \vec{v}}{\partial z}\delta z + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \vec{v}\,[x,y,z,t] + \frac{\partial \vec{v}}{\partial x} v_x\,\delta t + \frac{\partial \vec{v}}{\partial y} v_y\,\delta t + \frac{\partial \vec{v}}{\partial z} v_z\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \vec{v}\,[x,y,z,t] + \left(v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z}\right)\vec{v}\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \vec{v}\,[x,y,z,t] + (\vec{v}\cdot\nabla)\vec{v}\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t
\end{aligned}$$

Then,

$$\delta\vec{v} \approx \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right]\delta t$$

or, to be more precise,

$$\delta\vec{v} = \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right]\delta t + \mathcal{O}[(\delta t)^2]$$

Thus, we have

$$\frac{d\vec{v}}{dt} = \frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}$$

Alternatively, here is the same derivation using index notation:

$$\begin{aligned}
\delta\vec{v} &\approx \frac{\partial \vec{v}}{\partial x_i}\delta x_i + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \frac{\partial \vec{v}}{\partial x_i} v_i\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \left(v_i\frac{\partial}{\partial x_i}\right)\vec{v}\,\delta t + \frac{\partial \vec{v}}{\partial t}\delta t \\
&= \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v}\right]\delta t
\end{aligned}$$

Note: Here it becomes clear why we choose to write $\vec{v}\cdot\nabla$. It is a bit trickier to define the gradient of a vector field, but since we are dotting it with the velocity anyways, we can think of $\vec{v}\cdot\nabla$ as one single operator given by

$$\vec{v}\cdot\nabla = v_x\frac{\partial}{\partial x} + v_y\frac{\partial}{\partial y} + v_z\frac{\partial}{\partial z} = v_i\frac{\partial}{\partial x_i}$$

4.3

(a) A velocity normal to the surface causes a change in volume, so for an infinitesimal surface element $d\vec{S}$, the rate of change in volume is given by $\vec{v}\cdot d\vec{S}$. Integrating, the net rate of change in volume is given by

$$\oint_S \vec{v}\cdot d\vec{S}$$

(b) This should be equal to the time derivative of the volume, so we have

$$\frac{d}{dt}\int_V dV = \oint_S \vec{v}\cdot d\vec{S}$$

Applying the Divergence Theorem, we have

$$\frac{d}{dt}\int_V dV = \int_V \nabla\cdot\vec{v}\,dV$$

(c) Taking an infinitesimal volume element δV, we can get rid of the integrals, giving

$$\frac{d}{dt}(\delta V) = \nabla \cdot \vec{v}\,\delta V$$

Thus,

$$\nabla \cdot \vec{v} = \frac{1}{\delta V}\frac{d\delta V}{dt}$$

4.4

(a) By product rule, and then *Exercise 4.3(c)*, we have

$$\frac{d}{dt}(\xi\,\delta V) = \xi\frac{d\delta V}{dt} + \delta V\frac{d\xi}{dt} = \xi\nabla \cdot \vec{v}\,\delta V + \delta V\frac{d\xi}{dt} = \left(\xi\nabla \cdot \vec{v} + \frac{d\xi}{dt}\right)\delta V$$

Here we see the product rule has a physical interpretation as well. We not only have to account for the change in ξ as the particle moves (given by the material derivative), but since it is a unit-volume quantity, we also have to account for the expansion/shrinking of the fluid element (given by the divergence term).

(b) Integrating over an arbitrary volume, we get

$$\frac{d}{dt}\int_V \xi\,dV = \int_V \left(\xi\nabla \cdot \vec{v} + \frac{d\xi}{dt}\right)dV$$

4.5

(a) Again by product rule, we have

$$\frac{d}{dt}(\varphi\rho\,\delta V) = \rho\,\delta V\frac{d\varphi}{dt} + \varphi\frac{d}{dt}(\rho\,\delta V)$$

However, $\rho\,\delta V$ is the mass of the infinitesimal fluid element which we are following. Remember we consider fluid elements with fixed mass, so it may expand or shrink over time, but the mass will not change. Thus, the time derivative disappears and we have

$$\frac{d}{dt}(\varphi\rho\,\delta V) = \rho\,\delta V\frac{d\varphi}{dt}$$

Unlike the previous exercise, a fixed mass means we only have to account for the particle motion with the material derivative.

(b) Again, integrating over an arbitrary volume, we get

$$\frac{d}{dt}\int_V \varphi\rho\,dV = \int_V \rho\frac{d\varphi}{dt}\,dV$$

Section 5

5.1

(a) $m = \int_V \rho\,dV$

(b) From *Exercise 4.4*, we have

$$\frac{dm}{dt} = \frac{d}{dt}\int_V \rho\, dV = \int_V \left(\rho \nabla \cdot \vec{v} + \frac{d\rho}{dt}\right) dV = 0$$

Since this holds for any arbitrary volume, we can take an infinitesimal volume, getting rid of the integral, giving

$$\rho \nabla \cdot \vec{v} + \frac{d\rho}{dt} = 0$$

Here we expand the material derivative, giving

$$\frac{\partial \rho}{\partial t} + (\vec{v} \cdot \nabla)\rho + \rho \nabla \cdot \vec{v} = 0$$

We can see that if we show $\nabla \cdot (\rho \vec{v}) = (\vec{v} \cdot \nabla)\rho + \rho \nabla \cdot \vec{v}$, then we're done. Let us show this using index notation:

$$\begin{aligned} \nabla \cdot (\rho \vec{v}) &= \nabla_i (\rho v_i) \\ &= \rho \nabla_i v_i + v_i \nabla_i \rho \\ &= \rho \nabla \cdot \vec{v} + (\vec{v} \cdot \nabla)\rho \end{aligned}$$

Thus, we have the continuity equation

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

(c) Assuming incompressibility ($\rho =$ const.), we have $\nabla \cdot \vec{v} = 0$ (a.k.a the incompressibility condition).

5.2

(a) The mass flux is given by

$$\oint_S \rho \vec{v} \cdot d\vec{S}$$

(b) The time derivative of the mass should be the same as the mass flux. However, since a positive flux is a decrease in mass, we need to add a negative sign. Thus, we have

$$\frac{\partial}{\partial t}\int_V \rho\, dV = -\oint_S \rho \vec{v} \cdot d\vec{S}$$

Note that we use a partial derivative to clarify that we are working in the Eulerian perspective. Then, applying the divergence theorem and moving everything to one side gives us

$$\frac{\partial}{\partial t}\int_V \rho\, dV + \int_V \nabla \cdot (\rho \vec{v})\, dV = 0$$

(c) Since we are working with a fixed region, the spatial coordinates when we integrate aren't time dependent, so we can directly move the derivative inside the integral. Then, taking an infinitesimal volume element, we again get the differential form of the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{v}) = 0$$

5.3

(a) $\vec{p} = \int_V \vec{v}\rho \, dV$

(b) From *Exercise 4.5*, we have

$$\int_V \vec{f}\, dV = \vec{F} = \frac{d\vec{p}}{dt} = \frac{d}{dt}\int_V \vec{v}\rho \, dV = \int_V \rho\frac{d\vec{v}}{dt}\, dV$$

Taking an infinitesimal volume, we have

$$\rho\frac{d\vec{v}}{dt} = \vec{f}$$

Then, expanding the material derivative and dividing by ρ gives the momentum equation:

$$\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = \frac{\vec{f}}{\rho}$$

5.4

(a) Let us use index notation for readability. We have $\nabla p = \hat{\mathbf{x}}_i \nabla_i p$ and we can write $\nabla_i p = \nabla\cdot(p\hat{\mathbf{x}}_i)$. Thus,

$$\int_V \nabla p \, dV = \hat{\mathbf{x}}_i \int_V \nabla\cdot(p\hat{\mathbf{x}}_i)\, dV$$

Then, by divergence theorem,

$$\hat{\mathbf{x}}_i \int_V \nabla\cdot(p\hat{\mathbf{x}}_i)\, dV = \hat{\mathbf{x}}_i \oint_S p\hat{\mathbf{x}}_i \cdot d\vec{S} = \oint_S p(\hat{\mathbf{x}}_i\cdot d\vec{S})\hat{\mathbf{x}}_i = \oint_S p \, d\vec{S}$$

(b) Thus, we have

$$\int_V \vec{f}\, dV = \vec{F} = -\int_V \nabla p \, dV$$

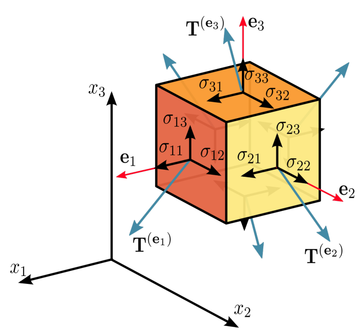
Since this is true for any arbitrary volume, we must have $\vec{f} = -\nabla p$ (this is ignoring long-range forces, which we can always add back later), and we can modify the momentum equation to get the Euler equation:

$$\frac{\partial\vec{v}}{\partial t} + (\vec{v}\cdot\nabla)\vec{v} = -\frac{\nabla p}{\rho}$$

Section 6

6.1

(a) σ_{ij} can be interpreted as the stress in the x_i through a surface with normal in the x_j. The normal stresses are σ_{11}, σ_{22}, and σ_{33}, and the other entries are the tangential stresses. This is why you often see this diagram for the stress tensor: (Credit: Wikipedia)



(b) Using multilinearity, we have

$$\sigma(\vec{v}, \vec{w}) = v_1\sigma(\hat{\mathbf{x}}_1, \vec{w}) + v_2\sigma(\hat{\mathbf{x}}_2, \vec{w}) + v_3\sigma(\hat{\mathbf{x}}_3, \vec{w}) = v_i\sigma(\hat{\mathbf{x}}_i, \vec{w})$$

Then, we can do the same thing to expand $\vec{w}$, giving

$$\sigma(\vec{v}, \vec{w}) = v_i\sigma(\hat{\mathbf{x}}_i, \hat{\mathbf{x}}_j)w_j = v_i\sigma_{ij}w_j$$

(c) i. We can rewrite $\sigma_{ij}w_j$ as $\sigma\vec{w}$ using the rules of matrix-vector multiplication (remember j is being summed over), which collects $\sigma_{ij}w_j$ for each i as the components of the resulting vector:

$$\sigma\vec{w} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} \sigma_{11}w_1 + \sigma_{12}w_2 + \sigma_{13}w_3 \\ \sigma_{21}w_1 + \sigma_{22}w_2 + \sigma_{23}w_3 \\ \sigma_{31}w_1 + \sigma_{32}w_2 + \sigma_{33}w_3 \end{bmatrix} = \begin{bmatrix} \sigma_{1j}w_j \\ \sigma_{2j}w_j \\ \sigma_{3j}w_j \end{bmatrix}$$

So $(\sigma\vec{w})_i = \sigma_{ij}w_j$.

ii. From part (a), we can interpret $1 \cdot \sigma_{ij} \cdot w_j$ as the stress in the x_i direction through a surface with normal $\vec{w}$ (again j is summed over). Thus $\sigma\vec{w}$ can just be interpreted as the net stress vector (force per unit area) through a surface with normal $\vec{w}$. This means we can also interpret the stress tensor as a linear transformation $\sigma : V \to V$ that takes in the normal to the surface, and gives a stress vector back.

iii. We have $v_iu_i = \vec{v} \cdot \vec{u}$. Since $\vec{u}$ is the net stress, if we want the stress in a specific direction $\vec{v}$ (let's assume it's a unit vector), we can just project $\vec{u}$ onto the direction by taking a dot product.

6.2

(a) i. We have $T_{23} = R_{2p}R_{3q}T_{pq} = R_{21}R_{33}T_{13} = T_{13}$. Thus $T_{13} = -T_{13} = 0$ and also $T_{23} = 0$.

ii. We have $T_{31} = R_{33}R_{12}T_{32} = -T_{32}$ and $T_{32} = R_{33}R_{21}T_{31} = T_{31}$. Thus, $T_{31} = T_{32} = 0$.

iii. We have $T_{11} = R_{12}R_{12}T_{22} = T_{22}$.

(b) i. We have $T_{12} = R_{11}R_{23}T_{13} = -T_{13} = 0$. Similarly, $T_{21} = R_{23}R_{11}T_{31} = -T_{31} = 0$.

ii. We have $T_{22} = R_{23}R_{23}T_{33} = T_{33}$.

(c) Thus, all the entries not on the diagonal are 0 and we have $T_{11} = T_{22} = T_{33}$. By letting $T_{11} = \alpha$, we have

$$T_{ij} = \alpha\delta_{ij}$$

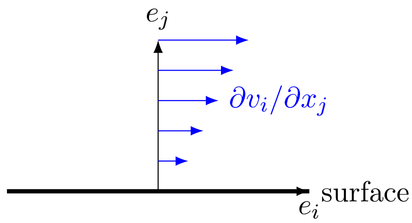
as desired.

Section 7

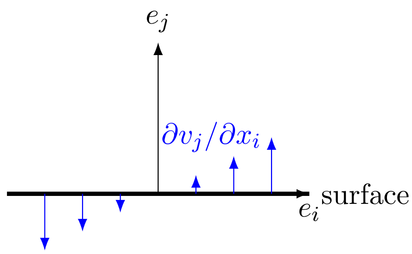
7.1 Since σ_{ij} represents the stress in the x_i direction a surface with x_j normal, naturally we might expect the derivatives to have indices of i and j, but let's make this intuition a bit more concrete.

For normal stresses (e.g. σ_{ii} where i is not summed over), part of it is due to pressure as we explained, but the other part it should be due to stretching along the direction of motion, meaning it should depend on derivatives with the same indices, like $\partial v_i/\partial x_i$, and it wouldn't make too much sense for it to depend on the other derivatives.

Recall the diagram of shear stress from *Section 2.1.* For shear stresses we expect two contributions. The first one is $\partial v_i/\partial x_j$, which represents velocity field in the x_i direction which varies along the x_j direction. This describes the sliding between fluid layers that we identify with friction in solids.



The second contribution is $\partial v_j/\partial x_i$, which describes velocity normal to the surface that vary along the surface. This tends to rotate the surface, and as we will see, viscosity is related to the deformation of the fluid.



7.2

(a) Consider a solid disk spinning with angular velocity $\vec{\Omega}$. The instantaneous velocity of a point on the disk at position $\vec{r}$ away from the center would also be given $\vec{\Omega} \times \vec{r}$. However, in the case of a solid disk, we think of there being no relative movement between particles, and we think of the disk moving together as a whole. Thus, intuitively, any shear stress should disappear in this case.

(b) We have $v_i = \epsilon_{imn}\Omega_m x_n$. Plugging this into the symmetric part, we get

$$\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} = \frac{1}{2}\left(\frac{\partial}{\partial x_l}\epsilon_{kmn}\Omega_m x_n + \frac{\partial}{\partial x_k}\epsilon_{lmn}\Omega_m x_n\right)$$

The derivative only affects x_n and we see that $\partial x_l/\partial x_n = 1$ when $l = n$ and 0 otherwise, which is exactly the Kronecker delta. Thus, we have

$$\begin{aligned}\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{sym}} &= \frac{1}{2}(\epsilon_{kmn}\Omega_m\delta_{ln} + \epsilon_{lmn}\Omega_m\delta_{kn})\\ &= \frac{1}{2}\Omega_m(\epsilon_{kml} + \epsilon_{lmk})\\ &= \frac{1}{2}\Omega_m(\epsilon_{kml} - \epsilon_{kml})\\ &= 0\end{aligned}$$

Here we used the antisymmetric property of the Levi-Civita where swapping any two indices causes the sign to reverse. We see that the antisymmetric part just changes the sign in the middle, so we must have

$$\left.\frac{\partial v_k}{\partial x_l}\right|_{\text{antisym}} = \frac{1}{2}\Omega_m(\epsilon_{kml} + \epsilon_{kml}) \neq 0$$

Since we argued that the viscous stress tensor should disappear under this motion, the antisymmetric part of cannot appear in it.

7.3

(a) Assuming the fluid is isotropic, we can write the rank-4 tensor as

$$A_{ijkl} = \xi' \delta_{ij}\delta_{kl} + \eta\delta_{ik}\delta_{jl} + \mu\delta_{il}\delta_{jk}$$

However, since we argued that the stress tensor is symmetric, meaning $\sigma_{ij} = \sigma_{ji}$, we must have $A_{ijkl} = A_{jikl}$, and we must have $\eta = \mu$. We keep η, leaving us with only two coefficients:

$$A_{ijkl} = \xi' \delta_{ij}\delta_{kl} + \eta(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})$$

In fact it is possible to show the stress tensor is symmetric in general ($\sigma_{ij} = \sigma_{ji}$), but we already have a symmetry here that we can use.

(b) Plugging this back into the viscous stress tensor, we have

$$\begin{aligned}
\sigma'_{ij} &= \left[\xi'\delta_{ij}\delta_{kl} + \eta(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})\right] \frac{1}{2}\left(\frac{\partial v_k}{\partial x_l} + \frac{\partial v_l}{\partial x_k}\right) \\
&= \xi'\delta_{ij}\frac{\partial v_k}{\partial x_k} + \eta\left[\frac{1}{2}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + \frac{1}{2}\left(\frac{\partial v_j}{\partial x_i} + \frac{\partial v_i}{\partial x_j}\right)\right] \\
&= \xi'\delta_{ij}\theta + \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)
\end{aligned}$$

(c) Substituting $\xi' = \xi - \frac{2}{3}\eta$, we have

$$\begin{aligned}
\sigma'_{ij} &= \left(\xi - \frac{2}{3}\eta\right)\delta_{ij}\theta + \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) \\
&= \eta\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} - \frac{2}{3}\delta_{ij}\theta\right) + \xi\delta_{ij}\theta
\end{aligned}$$

The trace is just given by σ'_{ii} (remember i is summed over):

$$\begin{aligned}
\operatorname{tr}\sigma'_{ij} &= \sigma'_{ii} \\
&= \eta\left(\frac{\partial v_i}{\partial x_i} + \frac{\partial v_i}{\partial x_i} - \frac{2}{3}\delta_{ii}\theta\right) + \xi\delta_{ii}\theta \\
&= \eta\left(\theta + \theta - \frac{2}{3}\cdot 3\theta\right) + \xi\cdot 3\theta \\
&= 0 + 3\xi\theta
\end{aligned}$$

As expected, the part with η is traceless, while the part with ξ contains the trace.

Section 8

8.1

(a) As we argued earlier, $\sigma\hat{n}$ or $\sigma_{ij}n_j$ gives the stress vector. Multiplying this by the area dS gives the force on that surface element $\sigma_{ij}n_j\,dS$ or $\sigma\cdot d\vec{S}$.

(b) Thus, the net force on the surface S is given by

$$F_i = \oint_S \sigma_{ij} n_j \, dS = \int_V \nabla_j \sigma_{ij} \, dV$$

or in vector form,

$$\vec{F} = \int_V \nabla \cdot \sigma \, dV$$

(c) Since

$$\int_V \vec{f} \, dV = \vec{F} = \int_V \nabla \cdot \sigma \, dV$$

for any volume V, we have $\vec{f} = \nabla \cdot \sigma$. Plugging this into the general momentum equation, we get

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = \frac{1}{\rho} \nabla \cdot \sigma$$

or

$$\frac{\partial v_i}{\partial t} + (\vec{v} \cdot \nabla) v_i = \frac{1}{\rho} \nabla_j \sigma_{ij}$$

which makes it clearer how we take the "divergence" of a rank-2 tensor.

8.2

(a) Taking the divergence of the stress tensor and assuming η and ξ to be constant, we have

$$\begin{aligned}
\nabla_j \sigma_{ij} &= \frac{\partial}{\partial x_j} \left[-p\delta_{ij} + \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_i}{\partial x_j} - \frac{2}{3} \delta_{ij} \theta \right) + \xi \delta_{ij} \theta \right] \\
&= -\frac{\partial p}{\partial x_i} + \eta \left(\frac{\partial^2 v_i}{\partial x_j^2} + \frac{\partial}{\partial x_i} \frac{\partial v_j}{\partial x_j} - \frac{2}{3} \frac{\partial \theta}{\partial x_i} \right) + \xi \frac{\partial \theta}{\partial x_i} \\
&= -\frac{\partial p}{\partial x_i} + \eta \left(\nabla^2 v_i + \frac{\partial \theta}{\partial x_i} - \frac{2}{3} \frac{\partial \theta}{\partial x_i} \right) + \xi \frac{\partial \theta}{\partial x_i} \\
&= -\frac{\partial p}{\partial x_i} + \eta \nabla^2 v_i + \left(\xi + \frac{1}{3} \eta \right) \frac{\partial \theta}{\partial x_i}
\end{aligned}$$

Thus we have

$$\nabla \cdot \sigma = -\nabla p + \eta \nabla^2 \vec{v} + \left(\xi + \frac{1}{3} \eta \right) \nabla \theta$$

Plugging this back into the Cauchy momentum equation and moving ρ to the LHS, we get:

$$\rho \left[\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} \right] = -\nabla p + \eta \nabla^2 \vec{v} + \left(\xi + \frac{1}{3} \eta \right) \nabla \theta$$

which is the **Compressible Navier-Stokes Equation**.

(b) Assuming incompressibility (e.g. $\nabla \cdot \vec{v} = \theta = 0$) and defining $\nu = \eta / \rho$, we have

$$\frac{\partial \vec{v}}{\partial t} + (\vec{v} \cdot \nabla)\vec{v} = -\frac{\nabla p}{\rho} + \nu \nabla^2 \vec{v}$$

which is the **Incompressible Navier-Stokes Equation**!!!

Section 10

As there are already multiple examples in *Section 10.1*, and the value for the problems in this section comes from working through them yourself, the solutions will not go into as much detail.

10.1 We have

$$\begin{aligned}\nabla\cdot(\nabla\times\vec{F}) &= \nabla_i(\nabla\times\vec{F})_i\\ &= \nabla_i\epsilon_{ijk}\nabla_j F_k\end{aligned}$$

Again we will be making use of the antisymmetry of the Levi-Civita. Noting that the derivatives can be exchanged we have

$$\begin{aligned}\nabla\cdot(\nabla\times\vec{F}) &= \nabla_i\epsilon_{ijk}\nabla_j F_k\\ &= \nabla_j\epsilon_{ijk}\nabla_i F_k\\ &= -\nabla_j\epsilon_{jik}\nabla_i F_k\\ &= -\nabla_j(\nabla\times\vec{F})_j\\ &= -\nabla\cdot(\nabla\times\vec{F})\end{aligned}$$

Thus we must have

$$\nabla\cdot(\nabla\times\vec{F}) = 0$$

10.2

(a) Expanding $\vec{v}\times\vec{\omega}$, we have

$$\begin{aligned}(\vec{v}\times\vec{\omega})_i &= [\vec{v}\times(\nabla\times\vec{v})]_i\\ &= \epsilon_{ijk}\epsilon_{klm}v_j\nabla_l v_m\\ &= \epsilon_{kij}\epsilon_{klm}v_j\nabla_l v_m\\ &= (\delta_{il}\delta_{jm}-\delta_{im}\delta_{jl})v_j\nabla_l v_m\\ &= v_j\nabla_i v_j - v_j\nabla_j v_i\end{aligned}$$

Note that product rule gives

$$\frac{1}{2}\nabla_i v_j^2 = v_j\nabla_i v_j$$

This gives

$$\begin{aligned}(\vec{v}\times\vec{\omega})_i &= \frac{1}{2}\nabla_i v_j^2 - v_j\nabla_j v_i\\ &= \frac{1}{2}\nabla_i v^2 - (\vec{v}\cdot\nabla)v_i\end{aligned}$$

Thus, we have

$$\vec{v}\times\vec{\omega} = \frac{1}{2}\nabla v^2 - (\vec{v}\cdot\nabla)\vec{v}$$

(b) Substituting this back into the "curled" Euler equation, we get

$$\frac{\partial\vec{\omega}}{\partial t} = \nabla\times(\vec{v}\times\vec{\omega})$$

(c) We again expand the double cross product, giving

$$\begin{aligned}[\nabla \times (\vec{v} \times \vec{\omega})]_i &= \epsilon_{ijk}\epsilon_{klm}\nabla_j(v_l\omega_m) \\ &= (\delta_{il}\delta_{jm} - \delta_{im}\delta_{jl})\nabla_j(v_l\omega_m) \\ &= \nabla_j(v_i\omega_j) - \nabla_j(v_j\omega_i)\end{aligned}$$

Note the parentheses. We will now expand the expression using product rule, and we will also use $\nabla \cdot \vec{\omega} = \nabla \cdot (\nabla \times \vec{v}) = 0$:

$$\begin{aligned}\nabla_j(v_i\omega_j) - \nabla_j(v_j\omega_i) &= v_i\nabla_j\omega_j + \omega_j\nabla_j v_i - v_j\nabla_j\omega_i - \omega_i\nabla_j v_j \\ &= v_i\nabla \cdot \vec{\omega} + (\vec{\omega} \cdot \nabla)v_i - (\vec{v} \cdot \nabla)\omega_i - \omega_i\nabla \cdot \vec{v} \\ &= (\vec{\omega} \cdot \nabla)v_i - (\vec{v} \cdot \nabla)\omega_i - \omega_i\nabla \cdot \vec{v}\end{aligned}$$

Thus, we have

$$\nabla \times (\vec{v} \times \vec{\omega}) = (\vec{\omega} \cdot \nabla)\vec{v} - (\vec{v} \cdot \nabla)\vec{\omega} - \vec{\omega}(\nabla \cdot \vec{v})$$

(d) Plugging this back into the equation from part (b), we have

$$\frac{\partial \vec{\omega}}{\partial t} = (\vec{\omega} \cdot \nabla)\vec{v} - (\vec{v} \cdot \nabla)\vec{\omega} - \vec{\omega}(\nabla \cdot \vec{v})$$

Moving the $(\vec{v} \cdot \nabla)\vec{\omega}$ term to the LHS and using the definition of the material derivative, we have

$$\frac{d\vec{\omega}}{dt} = (\vec{\omega} \cdot \nabla)\vec{v} - \vec{\omega}(\nabla \cdot \vec{v})$$

Exercise 10.3

(a) Recall in the derivation of the Cauchy Momentum equation that we wrote the force as

$$\oint_S \sigma\hat{n}\, dS$$

We can similarly write the torque as

$$\oint_S \vec{r} \times \sigma\hat{n}\, dS$$

We can also write the angular momentum as

$$\int_V \vec{r} \times \vec{v}\rho\, dV$$

Then, Newton's 2nd Law gives

$$\oint_S \vec{r} \times \sigma\hat{n}\, dS = \frac{d}{dt}\int_V \vec{r} \times \vec{v}\rho\, dV$$

(b) For the LHS, we apply the divergence theorem, giving

$$\begin{aligned}\oint_S \epsilon_{ijk}x_j\sigma kln_l\, dS &= \int_V \frac{\partial}{\partial x_l}(\epsilon_{ijk}x_j\sigma_{kl})dV \\ &= \int_V \epsilon_{ijk}\left(\delta_{jl}\sigma_{kl} + x_j\frac{\partial \sigma_{kl}}{\partial x_l}\right)dV \\ &= \int_V \epsilon_{ijk}\left(\sigma_{kj} + x_j\frac{\partial \sigma_{kl}}{\partial x_l}\right)dV\end{aligned}$$

For the RHS, since we follow the fluid region, we can use the results of *Exercise 4.5* to show that

$$\frac{d}{dt}\int_V \epsilon_{ijk}x_jv_k\rho\, dV = \int_V \epsilon_{ijk}x_j\frac{dv_k}{dt}\rho\, dV$$

(c) Setting the two sides equal to each other, we have

$$\int_V \epsilon_{ijk}\left(\sigma_{kj} + x_j\frac{\partial \sigma_{kl}}{\partial x_l}\right) dV = \int_V \epsilon_{ijk} x_j \frac{dv_k}{dt}\rho\, dV$$

Moving the term with the derivative of the stress tensor the RHS, we have

$$\int_V \epsilon_{ijk}\sigma_{kj}\, dV = \int_V \epsilon_{ijk} x_j \left(\rho\frac{dv_k}{dt} - \frac{\partial \sigma_{kl}}{\partial x_l}\right) dV$$

However, the expression in the parentheses on the RHS must be 0 because it exactly matches the terms in the Cauchy Momentum equation. This equation is also true for any arbitrary portion of fluid, meaning we can get rid of the integral. This leaves us with

$$\epsilon_{ijk}\sigma_{kj} = 0$$

(d) Multiplying both sides by ϵ_{imn}, we have

$$\begin{aligned}
\epsilon_{imn}\epsilon_{ijk}\sigma_{kj} &= 0\\
(\delta_{mj}\delta_{nk} - \delta_{mk}\delta_{nj})\sigma_{kj} &= 0\\
\sigma_{nm} - \sigma_{mn} &= 0
\end{aligned}$$

Finally, we have

$$\sigma_{mn} = \sigma_{nm}$$

as desired.